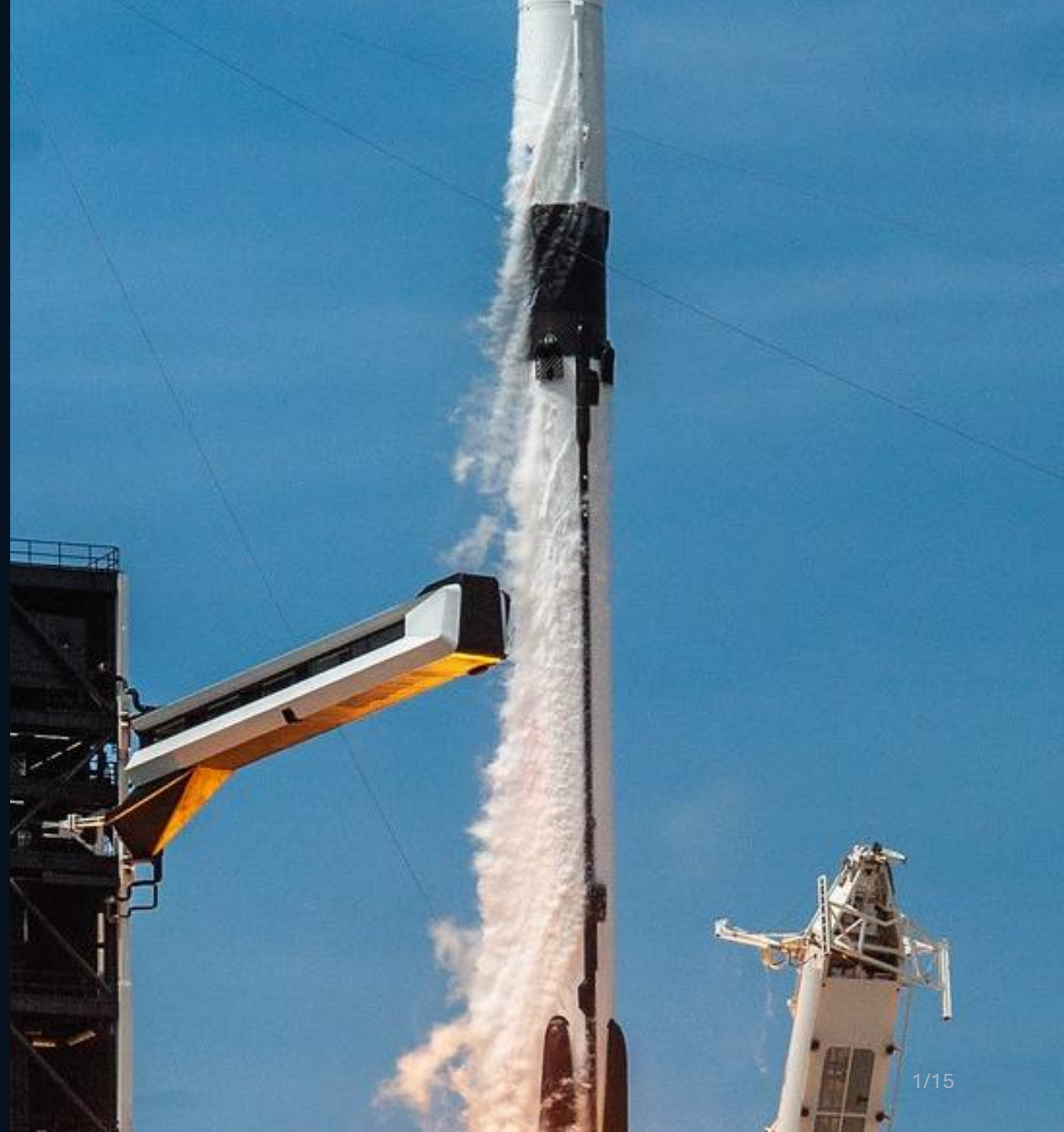


SPACE X FALCON 9

First Stage Landing Prediction

Data collection, EDA, SQL, interactive analytics, and classification modeling

<https://github.com/SinaQP/SpaceX-Falcon9-Prediction>



Executive Summary

90

cleaned launch records

66.7%

overall landing success

83.3%

best model test accuracy

87.7%

best CV score

- Goal: predict Falcon 9 first-stage landing success.
- Workflow: collect, wrangle, explore, map, dashboard, model.
- Best model: Decision Tree selected by cross-validation.
- Key drivers: launch site, orbit, reuse, grid fins, legs, and flight history.

Introduction

Business Context

- Reusable first stages reduce launch cost.
- Landing outcome affects launch economics.
- Prediction supports risk and cost estimation.

Project Question

- Can historical launch data predict landing success?
- Which operational features explain success patterns?
- How well do classification models perform?

Data Collection Methodology

- Collected SpaceX launch records and scraped Falcon 9/Falcon Heavy launch tables.
- Parsed fields: flight number, date, site, payload, payload mass, orbit, customer, launch and booster outcomes.
- Primary notebooks: web scraping and launch-data collection notebooks.
- Raw source file in repo: spacex_launch_data.csv.

Raw Data Fields

- Launch site
- Payload mass
- Orbit
- Customer
- Booster version
- Landing outcome

Data Wrangling Methodology

Step	Result
Clean records	90 rows and 18 columns retained for modeling
Target label	Class = 1 successful landing; Class = 0 unsuccessful landing
Missing values	PayloadMass filled with mean; LandingPad missingness retained as feature context
Core features	FlightNumber, Date, PayloadMass, Orbit, LaunchSite, Flights, GridFins, Reused, Legs, Block, ReusedCount

60

successful landings

EDA And Visual Analytics Methodology

- Explored landing success by launch site, payload mass, orbit, flight number, and year.
- Used pandas, matplotlib, seaborn, SQL, Folium, Plotly, and Dash.
- Built static charts for relationships and trends.
- Built interactive map and dashboard for site and payload exploration.

Static EDA	SQL	Folium	Dash
Trends and feature relationships	Query-based summaries	Launch-site geography	Interactive filters and plots

Predictive Analysis Methodology

Modeling Pipeline

- 83 engineered numerical and one-hot encoded features.
- StandardScaler applied before modeling.
- 80/20 train-test split with random_state=2.
- 10-fold GridSearchCV for hyperparameter tuning.
- Models: Logistic Regression, SVM, Decision Tree, KNN.

83

engineered features

20%

test split

EDA Visualization Results

Launch-site landing success



Success improved from early failures to 90.0% in 2019 and 84.2% in 2020.

- 2014-2015: 33.3%
- 2016: 62.5%
- 2017: 83.3%
- 2019: 90.0%
- 2020: 84.2%

EDA Feature Insights

Feature	False / No	True / Yes	Interpretation
Grid fins	10.0%	82.9%	Major landing-control signal
Landing legs	5.3%	83.1%	Strong readiness indicator
Reused booster	58.5%	78.4%	Reuse maturity correlates with success

Orbit examples: VLEO 85.7%, LEO 71.4%, ISS 61.9%, GTO 51.9%.

EDA With SQL Results

SQL Question	Result
Unique launch sites	CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40
NASA CRS total payload	45,596 kg
Average payload mass for F9 v1.1	2,928.4 kg
First successful ground-pad landing	2015-12-22
Top early landing outcome	No attempt: 10 between 2010-06-04 and 2017-03-20

Interactive Map Results Using Folium

Map Findings

- Launch sites plotted with Folium markers.
- Green markers = success; red markers = failure.
- Sites cluster near coastlines for launch safety and downrange paths.
- Distance markers compare sites with coastlines, highways, railways, and cities.

Geographic pattern

Florida sites cluster around Cape Canaveral / Kennedy Space Center. VAFB appears on the California coast.

Plotly Dash Dashboard Results

Launch Site	Launches	Success Rate	Dashboard Insight
KSC LC-39A	13	76.9%	Highest success
CCAFS SLC-40	7	42.9%	Mid performance
VAFB SLC-4E	10	40.0%	Mid performance
CCAFS LC-40	26	26.9%	Lowest success

Controls

- Site dropdown
- Payload slider
- Pie chart
- Payload scatter

Predictive Analysis Classification Results

Model	Test Accuracy	Best CV Score	Decision
Decision Tree	0.833	0.877	Selected
K Nearest Neighbors	0.833	0.848	Comparable test accuracy
Support Vector Machine	0.833	0.848	Comparable test accuracy
Logistic Regression	0.833	0.846	Lowest CV score

Decision Tree had the highest cross-validation score while matching the best test accuracy.

Innovative Insights

- Landing success is not just a payload-mass problem; hardware configuration is highly predictive.
- Grid fins and landing legs separate low-success early profiles from mature landing profiles.
- Reused boosters show higher success in the cleaned dataset, suggesting operational learning and reuse maturity.
- The clearest strategic signal is time: landing reliability changed sharply after 2015.
- Launch site matters because each site represents different missions, trajectories, and operating constraints.

Conclusion

90

cleaned records

66.7%

landing success

83.3%

test accuracy

Tree

selected model

- Full workflow: collected, cleaned, explored, mapped, dashboarded, and modeled Falcon 9 outcomes.
- EDA showed success improved over time and varied by site, orbit, and booster configuration.
- Folium and Dash made location and payload patterns easier to inspect interactively.
- Historical launch, mission, and booster features predict landing success with meaningful accuracy.